



SWOT Report

Yummy Future

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Yummy Future

TABLE OF CONTENTS

Introduction.....	4
Strengths	5
Strategic Highlights.....	5
Technological Innovation and Automation	5
Operational Efficiency and Consistency	6
Sustainability and Distinct Brand Positioning.....	6
Weaknesses	7
Strategic Highlights.....	7
Operational Fragility from Automation	7
Lack of Human Interactions	7
Drink Quality Concerns	8
Limited Appeal for Health-Conscious Students	9
Ethical Concerns and Negative Perceptions around Job Loss	9
Opportunities	11
Strategic Highlights.....	11
High Demand and Technology-Driven Demographic.....	11
Innovative and Unique Design	12
Precision, Accuracy, and Speed.....	12
Competitive Advantage and Sustainability Positioning.....	12
Threats	13
Strategic Highlight	13
Tariffs & instability in coffee supply chain	13
Reduced Disposable Income.....	13
Climate Change Effects on Coffee Bean Production.....	13
Competition from Traditional Cafes.....	14
Technical Failures	14
Health Perceptions	14
Ethical Issues of Automation & Robotic Labor.....	15

Recommendations	16
Increase Humanlike Feature to Improve Trust & Engagement	16
Improve Marketing on Sustainability & Speed	16
Expand Menu to Include Health-Conscious Choices	17
Build Preventive Maintenance Backup Protocols	17
Work Around Supply Chain Instability.....	17
Build a stronger Brand Identity Around Experience.....	17
Conclusion	18
Works Cited.....	19

Introduction

Yummy Future is a robotic coffee shop founded by UIUC alumni, built on automation, AI-driven robotics, and sustainability-centered innovation. As one of the few fully robotic cafes in the Midwest, Yummy Future operates in a largely under-researched area within hospitality technology. Existing information on service robots is limited and often focused more on hotels and restaurants than university-based robotic coffee shops (Belanche et al., 2021).

This SWOT report will take a closer look at Yummy Future, exploring how cafe automation enhances campus life. While there isn't a significant amount of secondary research on robotic cafes, our own primary research is new and assists in filling a gap, offering insight into how Yummy Future functions, where its strengths lie, and its potential for growth. By exploring the robotic cafe, we can identify potential challenges and threats that existing research hasn't covered. Overall, the report offers a clear and up-to-date view of Yummy Future's current position as well as its future potential.

This report identifies the unique strengths that set Yummy Future apart from traditional cafes, such as its advanced technology used to craft drinks, fast service model, and ability to attract a specific audience who are attracted to their innovation. Additionally, it identifies the weaknesses that have the potential to hinder the businesses growth, including reliability concerns, occasional inconsistencies in product quality, and technological challenges that would come with relying heavily on advanced technology in a busy environment.

We also identify and highlight the opportunities Yummy Future can take advantage of as robotic cafes continue to grow in popularity. From expanding their menu to building stronger brand awareness on and off campus, there are many ways the company could expand as interest in this innovation continues to rise. Finally, the report looks at potential threats that Yummy Future could face, such as having competitors, shifting consumer expectations, and the possibility of technological malfunctions.

By combining our original research with an insightful evaluation of internal and external factors, this SWOT analysis provides a thorough understanding of where Yummy Future stands today and where it can go in the future.

Our research draws on:

- Primary data: Website and Customer survey of UIUC students
- Secondary Data: Articles and research on service robots, customer psychology, ethics, sustainability, and the coffee supply chain

Strengths

STRATEGIC HIGHLIGHTS

Yummy Future benefits from several key strengths that position it as a competitive and forward-thinking business within the cafe industry. Its advantages extend across technology, operations, and brand identity, allowing the company to differentiate itself from traditional human-barista cafes and emerging competitors.

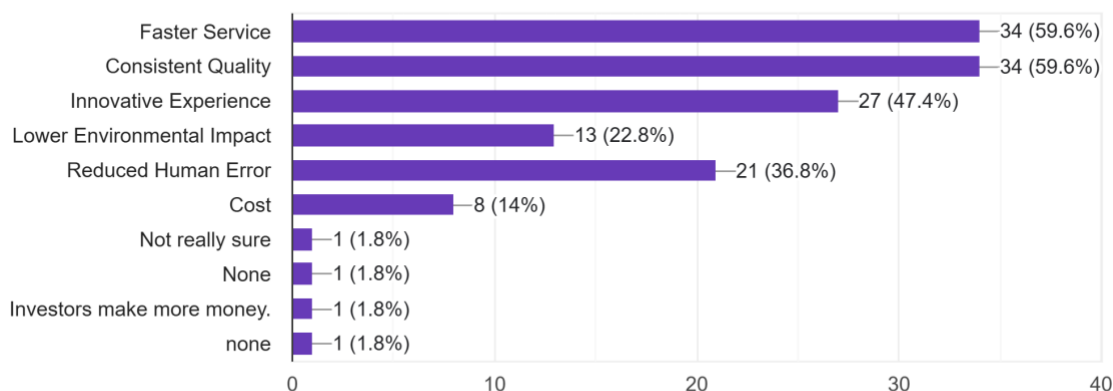
TECHNOLOGICAL INNOVATION AND AUTOMATION

One of Yummy Future's strongest advantages is its innovative robotic barista system, which creates a futuristic, automated experience that traditional cafes cannot replicate. Rather than treating automation as a hidden back-end tool, the cafe uses its robotic system as part of the customer experience. Customers can watch their drink being prepared through a transparent, automated process. According to our survey of 56 respondents, 46.4% selected "Innovative Experience" as a major advantage of automated cafes. Also, robotic units provide a consistent and novel customer experience that enhances brand identity. Out of 43 responses, 41.9% and 39.5% cited a unique, futuristic atmosphere and curiosity toward the automation process and as a motivating factor for visiting.

Hospitality research by Belanche et al. (2021) shows that when customers believe robots are introduced to enhance service quality, rather than simply reduce labor costs, they respond more positively and show higher willingness to return. This aligns with Yummy Future's branding and origin story—the company's mission emphasizing accessibility, high-tech service, and consumer-centered automation. These give Yummy Future a unique market position compared to conventional coffee shops like Starbucks or Espresso Royale.

What do you think are the main advantages of automated cafés like Yummy Future? (Check all that apply)

57 responses



Yummy Future

OPERATIONAL EFFICIENCY AND CONSISTENCY

Yummy Future's robotic system offers a level of efficiency and consistency that differentiates itself with other coffee shops. Many survey respondents highlighted "speed," "accuracy," and "no mistakes" as perceived strengths of robotic service. 26 out of 43 respondents, which is 60.5%, who had never visited an automated cafe, said they would be motivated by the speed of the service. Robotic preparation eliminates human error while maintaining standardized quality across visits.

Yummy Future also aligns with the hospitality research, which shows that the automated system improves reliability and reduces wait time, which creates an advantage during peak hours (Belanche et al., 2021), as the ordering kiosk machine and robot can handle continuous orders without fatigue. Human employees remain present for customer support and troubleshooting, allowing the cafe to combine robotic consistency with human oversight. This combination allows the cafe to be efficient without appearing impersonal.

SUSTAINABILITY AND DISTINCT BRAND POSITIONING

Another strength is Yummy Future's sustainability-oriented operations. Automated drink making ensures drinks follow the exact same recipe every time, results in reduced ingredients cost and wasted ingredients which ultimately leads to support cleaner and efficient workflow. From the investment report of Yummy Future, they made 300% margin profit by reducing labor cost and disproportionate use of ingredients due to its precise robot operations. In addition, several respondents from our survey noted that robotic preparation seems more sustainable due to "lower environmental impact", "less waste", and "energy efficient." This creates a distinct position in the beverage market: the cafe is not just a place to buy drinks, but a tech-forward experience that appeals to sustainability-minded customers.

Weaknesses

STRATEGIC HIGHLIGHTS

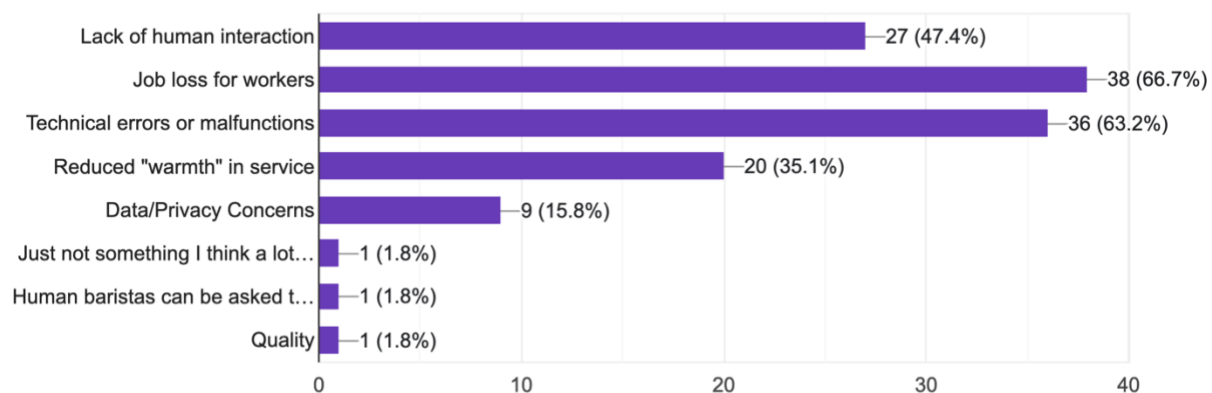
Although Yummy Future leverages automation as a strength, its dependence on robotics creates operational fragility, limits human interaction, raises drink quality concerns, narrows menu appeal, and generates unease about job loss.

OPERATIONAL FRAGILITY FROM AUTOMATION

Since Yummy Future depends on one robotic arm, any mechanical issue, such as a calibration error or software malfunction, may put the entire operation on hold. Although human staff are available, with no human barista present, even minor disruptions can lead to long delays. Whereas human-operated cafes can continue despite individual mistakes. Survey responses show that 64.3% of students consider "Technical errors or malfunctions" as a major concern.

What do you think are the main disadvantages or concerns about robotic cafés? (Check all that apply)

57 responses



Not only are malfunctions from the robotic service a concern, but people also hesitate due to a loss of trust in automated service. From our survey, many students expressed low trust in robotic service, often selecting "Don't fully trust automated services" and reporting lower trust/ satisfaction. These concerns included fear of malfunction and uncertainty about system reliability. These technical uncertainties and low customer trust in the automation service affect repeat visits and brand credibility.

LACK OF HUMAN INTERACTIONS

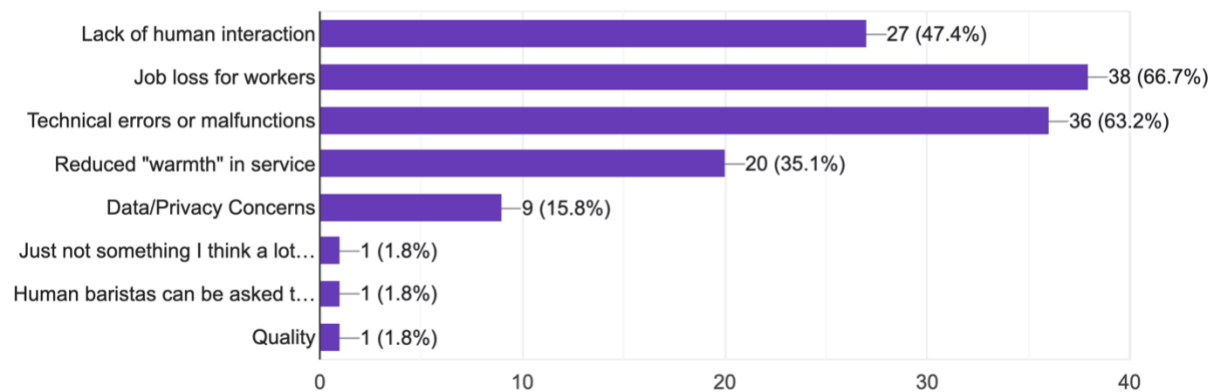
Without a human barista, Yummy Future offers limited interpersonal engagement. Students who value friendliness, conversations, or personalized drink adjustments may feel that robot baristas making their coffee are impersonal and less welcoming than traditional coffee shops. According to our survey, 48.2% of respondents selected "Lack of human interaction" as the main disadvantages of robotic cafés. One of

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our respondents emphasized in our survey that human baristas can handle unusual requests, make flexible customer service decisions, and prioritize customer satisfaction in a way robots cannot. Research supports this: customers value warmth, empathy, and social interaction in hospitality. Anthropomorphism studies show that trust increases when robots appear more human-like (Service robots & customer experience, 2025), which is something that Yummy Future's mechanical robots do not offer.

What do you think are the main disadvantages or concerns about robotic cafés? (Check all that apply)

57 responses



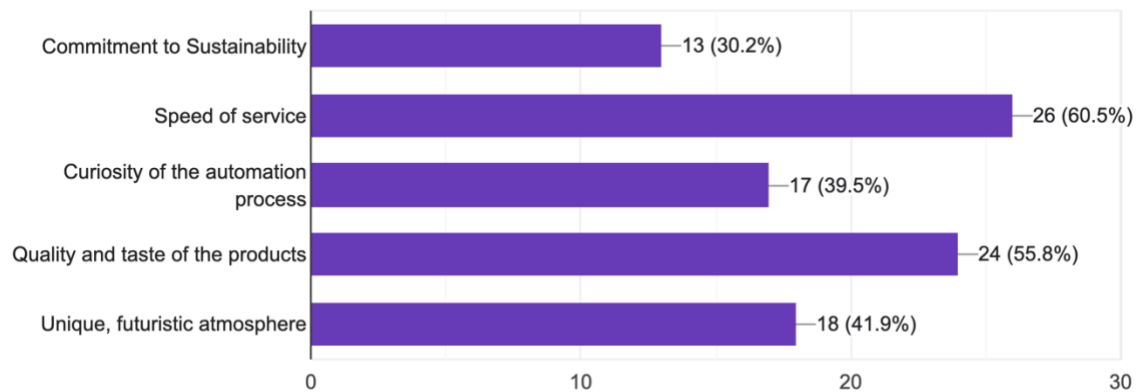
DRINK QUALITY CONCERNS

Robot-made drinks can lead to differences in quality of drinks. 58.7% of respondents indicated that they were hesitant to visit an automated cafe because they were "Unsure about the quality of drink". Robots follow preprogrammed recipes, which limits personalization and reduces menu flexibility. When a cafe has human baristas, they could receive feedback or personal drink adjustments that can indicate the need for change in the taste of their drinks, while robots cannot.

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What factors would motivate you to visit Yummy Future, the robotic café? (Check all that apply)

43 responses

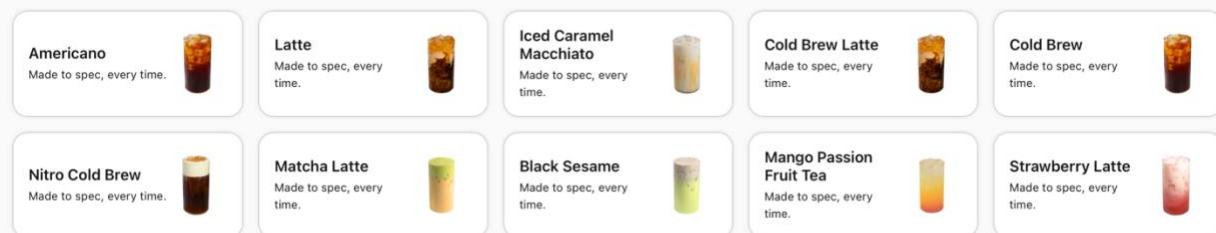


LIMITED APPEAL FOR HEALTH-CONSCIOUS STUDENTS

Yummy Future’s menu currently focuses primarily on caffeinated drinks and fast service. As more students attempt to limit caffeine intake due to anxiety, sleep issues, or health concerns, the lack of decaf or lower caffeine alternatives may limit appeal. While they offer Matcha drinks and fruit juice, they are very sweet. Also, there is no option for decaf or less caffeinated options for coffee. These limitations become more prominent as caffeine consumption patterns shift across college populations.

Drinks We Serve

Hot & iced classics, teas, and signatures.



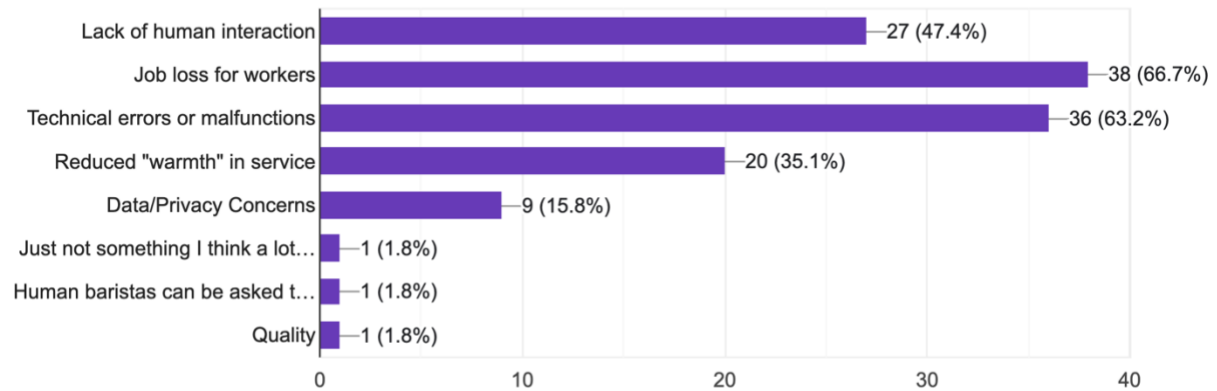
ETHICAL CONCERNS AND NEGATIVE PERCEPTIONS AROUND JOB LOSS

66.1% of our survey respondents selected “Job loss for workers” as a disadvantage, explicitly expressing discomfort with automation replacing human labor. Hospitality robotics research shows that automation can reduce job availability, undermine local labor economies, and shift responsibility from humans to machines (**Service Robots & Ethics, EHL, n.d.**). Students described ethical unease, which can harm long-term brand perception and discourage socially conscious customers.

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What do you think are the main disadvantages or concerns about robotic cafés? (Check all that apply)

57 responses



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Opportunities

STRATEGIC HIGHLIGHTS

Yummy Future can use high demand, a tech-savvy demographic, and its innovative robotic systems to expand and grow through precision, sustainability, and scalability.

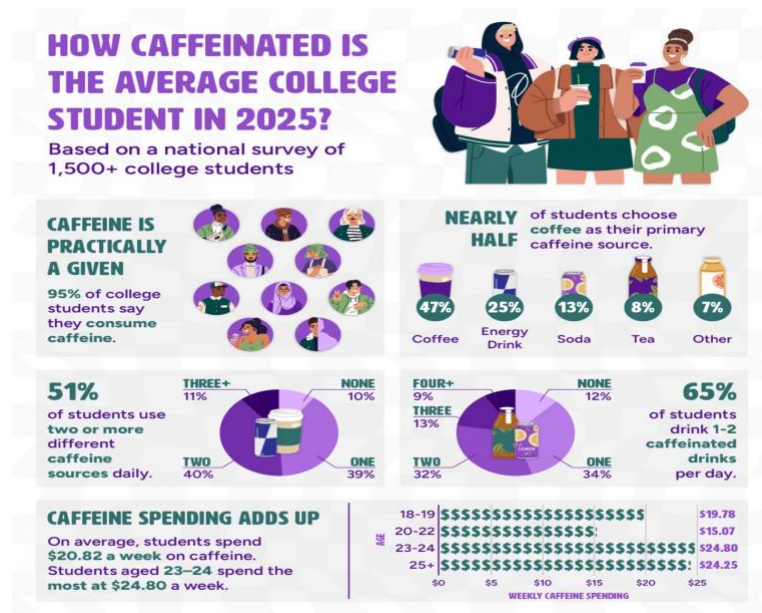
HIGH DEMAND AND TECHNOLOGY-DRIVEN DEMOGRAPHIC

The coffee shop being located in a college town ensures that the demand for the product is consistently high due to the fact that college students on average tend to have a high coffee consumption. Post-COVID studies show that 36% of students increased their coffee consumption (Coffee consumption after COVID-19, PMC11053568). This will help the coffee shop thrive due to the fact that they will always be in demand and can cater to a constant need.

Since Yummy Future is located in a college town, this ensures consistently high demand, as students are heavy coffee consumers. College students are also extremely familiar with technology and are open to interacting with robots, whether ordering drinks or working alongside automated systems. This makes the automation and robotic services at Yummy Future more novel and widely accepted. Recent research in Anthropomorphism & Customer Experience, 2025 also shows increasing positive attitudes toward robotic services when:

- Robots are efficient
- They reduce wait times
- They contribute to sustainability

This matches the campus culture at UIUC, where students value innovation, efficiency, and convenience.



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INNOVATIVE AND UNIQUE DESIGN

Incorporating robots in the service industry is a new and emerging concept throughout the world, but Yummy Future stands out as the only cafe on the UIUC campus where a robot assists and makes the drinks. This unique integration of technology not only differentiates Yummy Future from other coffee shops but also creates a novel experience that attracts customers, especially students, interested in innovation.

PRECISION, ACCURACY, AND SPEED

Having a robot regularly assist in making the drinks ensures consistent drink quality by precisely following a preprogrammed recipe that lowers variability that occurs with human baristas. This consistency can increase drink accuracy, reduce waste, and lower human error. Robots can also produce drinks at a faster pace than human baristas, reducing wait time during peak hours and improving customer satisfaction. All of this increases Yummy Future's reliability which encourages repeat visits, and increases the chances of positive customer reviews and recommendations. Over time, this can establish Yummy Future as a dependable and innovative choice for regular students and new customers looking for a tech-driven cafe experience.

COMPETITIVE ADVANTAGE AND SUSTAINABILITY POSITIONING

With rising minimum wages and ongoing labor shortages in the food service industry, robotic cafes like Yummy Future are more cost-efficient, allowing the business to be profitable while reducing dependence on human labor.

At the same time, Gen Z consumers prioritize sustainability and ethical consumption, creating an opportunity for Yummy Future to strengthen its brand by highlighting their environmentally conscious practices, like reducing waste, improving energy efficiency, maintaining a smaller carbon footprint, and using technological innovation to optimize resource use. Yummy Future has positioned itself for long-term profitability with the growing trend of "eco-tech" cafes.

Threats

STRATEGIC HIGHLIGHT

Yummy Future faces multiple threats including supply chain instability and rising coffee costs, competition from traditional cafes, technical and operational failures, shifting health trend, and ethical concerns surrounding automation.

TARIFFS & INSTABILITY IN COFFEE SUPPLY CHAIN

The United States imports over \$8.2 billion of coffee annually according to STiR Coffee & Tea Magazine, n.d., making it one of the world's largest import-dependent coffee markets. Because the US does not domestically produce a lot of coffee, any international trade policy shift significantly impacts US cafes. When tariffs are applied to coffee beans or related inputs (machinery, filters), costs rise at every stage:

- Importers face higher duties
- Roasters pay more for raw beans
- Retailers must raise prices or suffer losses.

According to Coffee Digest, n.d., tariffs are essentially an “indirect tax” on businesses that are paid by the customers. For a student-heavy customer base, who already are financially cautious, higher coffee prices may make frequent cafe visits less affordable. This can push consumers towards cheaper alternatives like instant coffee or energy drinks. For Yummy Future, where pricing is an important factor in attracting students, tariff-driven cost increases could weaken demand.

REDUCED DISPOSABLE INCOME

Macroeconomic instability, such as inflation, slower wage growth, or rising living expenses, directly affects consumer behavior. College students in particular face financial limitations from tuition, rent, groceries, and daily living expenses. Research shows that during an economic downturn, spending on non-essential items declines and consumers prioritize necessities (Socioeconomic consumption patterns, arXiv 2023). Food and beverages purchases become more price-sensitive.

In the cafe industry, premium drinks and specialty beverages are the first category consumers cut back on. Students limit cafe visits and choose cheaper coffee options to save money. Since Yummy Future's main consumer base is students at UIUC, any reduction in disposable income could reduce sales volume. Even slight price increases, due to supply chain costs or inflation, could make students switch to more affordable options.

CLIMATE CHANGE EFFECTS ON COFFEE BEAN PRODUCTION

Coffee is one of the most climate-sensitive crops in the world. Rising temperature and unpredictable rainfall threaten coffee-producing regions. Scientific studies project (Climate Change & Coffee Production, 2023) predict:

- Reduced arable land for coffee by 2050
- Lower crop yield due to heat stress
- Increased volatility in global coffee prices

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- Risk of long-term shortages

For cafes like Yummy Future that depend on consistent, high-quality bean supply, climate-driven problems represent a long-term threat. As bean scarcity increases, prices could increase, supply contracts can become unstable, and roasters may struggle to maintain quality. Even automated cafes cannot do anything about rising raw-material costs or declining bean availability.

COMPETITION FROM TRADITIONAL CAFES

While Yummy Future differentiates itself through robotics and automation, traditional cafes still dominate the industry. Competitors such as Starbucks, Dunkin and BrewLab provide several advantages that appeal to customers:

- Human interaction
- Customization and Flexibility
- Atmospheric Appeal
- Brand familiarity and social reputation

Research on service robots shows that customers often prefer human warmth in service situations that involve conversation. Even students who enjoy the novelty of robotic coffee may still view traditional cafes as more comfortable, social, and reliable. This competition poses a threat as Yummy Future tried to grow in a market where customer loyalty is tied to emotional experience.

TECHNICAL FAILURES

Robotic cafes rely entirely on software, sensors, mechanical arms, mixers, and automated dispensing systems. Although efficient, these systems have reliability risks that human-operated cafes do not like software glitches that could cause the robot to malfunction or mechanical breakdowns like stuck a robotic joint. Any malfunction could stop the entire cafe's operation until someone fixes it. Traditional cafes can adapt around a broken machine or assign tasks to other baristas, but robotic cafes lack this flexibility. This could also in turn increase operational costs for repairs and maintenance and reduce foot traffic or customer trust.

HEALTH PERCEPTIONS

Although caffeine use among college students is very high, shifting health views may influence future consumption. Research by Mahoney et al. (2018) links excessive caffeine consumption to:

- Heightened anxiety
- Sleep disturbances
- Stress related symptoms
- Elevated blood pressure

Gen Z is very health-conscious, with interest rising in low-caffeine beverages, herbal beverages, and non-stimulated drinks. If health-related concerns grow, students might reduce consumption or switch to lower-caffeinated options that Yummy Future may not currently have.

ETHICAL ISSUES OF AUTOMATION & ROBOTIC LABOR

As automation use increases in the service industry, societal concerns grow around the ethics of replacing human labor with machines. Ethical criticisms revolve around:

- Job displacement
- Dehumanization of service
- Reduced community connection, as local cafes serve as social spaces
- Dependence on proprietary technology, raising worries about algorithmic bias
- Privacy concerns, especially if robots collect or store consumer data

Consumers who feel uneasy about automation might avoid robotic cafes out of principle, even if the service is efficient. For Yummy Future, whose identity is centered on automation, ethical scrutiny represents a long-term reputational risk.

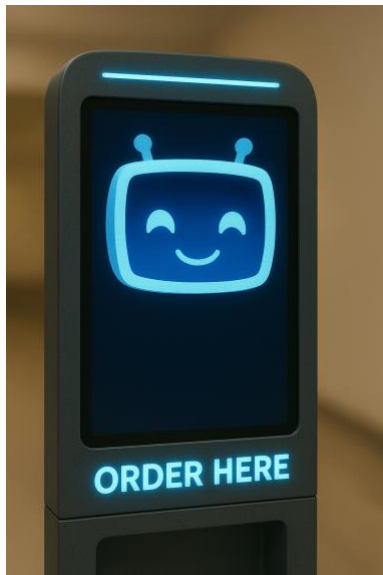
Recommendations

INCREASE HUMANLIKE FEATURE TO IMPROVE TRUST & ENGAGEMENT

Research in the service industry shows that customers feel more comfortable when service robots show small human-like movements and behaviors. These behaviors do not need to be realistic; even simple forms of human-like gestures can make interactions feel warmer and more intuitive. Yummy Future could integrate features such as:

- A friendly greeting voice that welcomes customers during ordering
- LED “faces” or expressive animations to simulate mood – IMAGES
- Short, conversational phrases (“Your drink is on the way!”)
- Fun animations when drinks are completed

Features like this would help reduce the gap between humans and machines, easing customer anxiety and garnering trust.



IMPROVE MARKETING ON SUSTAINABILITY & SPEED

Although Yummy Future has a competitive advantage in efficiency and sustainability, they are not widely recognized by students. By promoting benefits, the cafe can communicate the value of the robotic service. Potential marketing messages include:

- “Drinks prepared in 30 seconds - perfect for drinks on the go”
- “Sustainable robotics helps redefine eco-friendly cafes”
- “Up to X% less ingredient waste due to precision automation”
- “Energy efficient machines reduce environmental impact”

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EXPAND MENU TO INCLUDE HEALTH-CONSCIOUS CHOICES

To appeal to students with different dietary needs and preferences, the cafe should expand its menu by offering:

- Decaf coffee
- Lower-caffeine alternatives
- Reduces-sugar or sugar-free options
- Seasonal or functional beverages like herbal teas

This would align well with changing health trends among Gen Z.

BUILD PREVENTIVE MAINTENANCE BACKUP PROTOCOLS

Because robotic cafes rely on technology, system failures pose major operational risks. To reduce downtime and maintain customer trust, Yummy Future should implement preventative measures like:

- A trained rapid-response technician team on campus
- On-site storage of essential spare mechanical parts
- Remote troubleshooting capabilities for off-hours issues
- Real-time diagnostic software that alerts staff to issues before they escalate

This would ensure that the robotic systems

WORK AROUND SUPPLY CHAIN INSTABILITY

Since tariffs and climate-related risks threaten coffee availability and price, Yummy Future can:

- Have different suppliers from different regions
- Enter long-term contracts that lock in prices

These strategies could help maintain stable pricing and protect the cafe from instability.

BUILD A STRONGER BRAND IDENTITY AROUND EXPERIENCE

Yummy Future's unique selling point is its novelty. Strengthening the brand identity can reinforce the "futuristic cafe experience" and deepen brand loyalty. The brand can be strengthened through:

- Social media storytelling
- Collaborations with student influencers
- Loyalty programs

Conclusion

In conclusion, Yummy Future demonstrates a unique position as one of the few fully robotic cafes in the Midwest. Its combination of automation, AI-driven robotics, and overall focus on sustainability allows it to stand out from traditional coffee shops, and its fast service and technology leading approach attract a specific audience that has an interest in innovation. These significant strengths not only make the cafe interesting for students and visitors but also show how technology and AI can impact the expectations we have for coffee and service as a whole.

Regardless of Yummy Future's significant strengths, there are also significant challenges that should be considered. Because the cafe relies so heavily on technology, the possibility of malfunctions, inconsistencies in drink quality, and other reliability issues remain. Additionally, because there aren't much significant research or examples of robotic cafes on university campuses, Yummy Future is establishing a new concept. This entails the need for the team to address issues and complications as they arise and refine the system continuously to ensure a positive experience for customers.

Because Yummy Future is a relatively new concept, there remain several opportunities for growth. Yummy Future has the opportunity to grow by leveraging high demand, a tech-savvy campus student body, and its innovative robotic systems. By focusing on precision, sustainability, and scalability, the cafe can expand its reach, serve more customers efficiently, and continue standing out as a leader in robotic service.

At the same time, Yummy Future faces several threats, including supply chain instability and rising coffee costs, competition from traditional cafes, potential technical or operational failures, shifting health trends, and ethical questions surrounding automation. The key will be balancing its strengths and opportunities with careful planning and significant risk management. Overall, this SWOT analysis shows that Yummy Future has strong potential to be the leader in showing how technology, innovation, and sustainability can shape the future of customer service.

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